

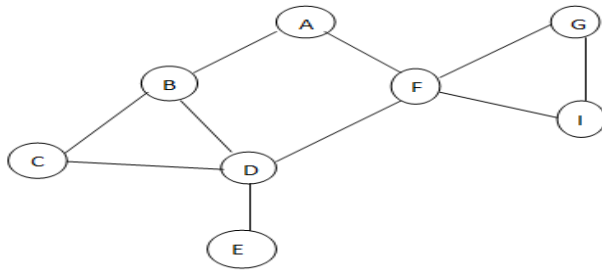
Master of Computer Applications (MCA)
CS-409: Design and Analysis of Algorithms
Mid Exam - Semester: II
March 2024

Time: 1.5 Hours

Max. Marks: 30

Attempt all questions. Parts of a question should be answered together.

1. a. Arrange the following functions in the increasing order of their rate of growth: 2
$$n^{\lg n}, \quad \sqrt{n}, \quad n \log n, \quad n, \quad \frac{100}{n}, \quad n^{\sqrt{n}}$$
- b. Consider the Quicksort algorithm. Suppose there is a procedure for finding a pivot element which splits the list into two sub-lists each of which contains at least one-fifth of the elements. Let $T(n)$ be the number of comparisons required to sort n elements. Then find the recurrence relation and time complexity for $T(n)$. 4
2. a. Consider a sorting algorithm A with run time $O(n)$. What is the condition on A for it to be usable as the intermediate sort in Radix sort? Is it possible to use quick sort as intermediate sort? Justify your answer. 4
- b. Write a recursive algorithm for insertion sort. 4
3. a. A priority queue is implemented as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is: 10, 8, 5, 3, 2. Two new elements 1 and 7 are inserted into the heap in that order. What will be the level-order traversal of the heap after the insertion of the elements. 4
- b. Consider a sorted array of n numbers. Write an algorithm to find a pair a and b such that $|a-b| = k$, where k is a positive integer, and also calculate time complexity for the same. 4
- c. Run the Depth first search (DFS) algorithm and Breadth first search (BFS) algorithms on the network given in the figure starting from vertex $s=A$. Also assume that both the BFS and DFS algorithms will choose the left-most node first. 8



- i. Determine if the graph is bipartite. If yes, give the two partitions else justify your answer.
- ii. Find the tree edges, cross edges and back edges using DFS.
- iii. Explain how DFS can be used to find a cycle in the graph
- iv. Given a pair of vertices s and t , explain how BFS can be used to find a shortest path between s - t in graph.

Master of Computer Applications (MCA)
CS-409: Design and Analysis of Algorithms
Semester-II
May 2024

Time: 3 Hours

Max Marks: 60

Instruction: Attempt all questions. Parts of a question should be answered together.

1. (a) What is the worst-case number of arithmetic operations performed by recursive binary search on the sorted array of size n ? consider the above scenario, compute the time complexity. 3
- (b) Write a compute prefix function for given pattern using KMP. Also apply the same on the following pattern P: ABCDABCAABBCCABCDAB 3
- (c) Consider an array representation of binary min-heap containing 1023 elements. How many numbers of comparisons (in best-case) required to find the maximum element in the heap. 4

2. (a)

```
#include<stdio.h>
Int fun(int x, int y)
{
    x=2*x+y;
    return x;
}
int main()
{
    int x=2, y=5;
    y=fun(y, x);
    x=fun(y, x);
    printf("%d\n", x);
}

```

Find the value of x . 3
- (b) Give the three best case instances for insertion sort so that it takes the atmost $n-1$ number of comparisons. 3
- (c) Consider the following undirected graph G: 4

(A,B)=4, (B,C)=4, (A,C)=1, (A,D)=3, (A,E)=x, (D,E)=5, (C,D)=4
 Find the value of x for the same and draw the all possible MWSTs.

3. (a) Consider the recurrence function 5

$$T(n) = \begin{cases} 2T(\sqrt{n}) + \log n & \text{if } n > 2 \\ 2 & \text{if } 0 < n \leq 2 \end{cases}$$

Find the $T(n)$ in terms of θ notation.

5

- (b) Consider the following directed graph.

(A,B), (B,C), (C,F), (A,D), (D,E), (E,F)

How many numbers of topological orderings of the vertices is/are possible.

4. (a) Let Q be a quicksort algorithm to sort numbers in ascending order using last element as the pivot. Let C_1 and C_2 be the number of comparisons made by Q for inputs $[1, 2, 3, 4, 5]$ and $[4, 1, 5, 3, 2]$ respectively. Find the values of C_1 and C_2 (As discussed in the class). 5

- (b) We define the Fibonacci numbers as follows: 5
 $F[0]=0$;
 $F[1]=1$;
 $F[n]=F[n-1]+F[n-2]$ for all $n>1$;
 Write a `Fib_Memoize(n)`. It should be recursive but it should memorize its answers so that it runs in time $O(n)$. Explain the `Fib_memoize` with recursive tree for $n=8$.

5. (a) Consider a weighted interval scheduling using dynamic programming, we assume that the intervals are sorted by their finishing time. Where s_i , f_i and v_i are the starting, finishing and weights of i^{th} activity respectively. Find maximum weights/values subset of mutually compatible activities (Two activities are compatible if they don't overlap each other). 7

i	1	2	3	4	5	6	7
s_i	1	3	0	5	3	5	6
f_i	4	5	6	7	9	9	10
v_i	3	2	4	1	3	5	2

Write a recursive formula for binary choice to select an activity and draw the recursive tree, and find the solution for the same. Is it possible to apply greedy algorithm on the same if yes give the solution otherwise state the reason.

- (b) An instructor has given a test to her/his class of n students. The maximum marks for the test is 100. The instructor decided not to give the fractional marks while grading the test. The instructor wishes to sort n integer scores in descending order. Design a linear algorithm to perform this task. Also list the assumptions (if any) that you make to solve the problem. 3
6. (a) Prove that lower bound of $\lceil \frac{3n}{2} \rceil - 2$ comparisons in worst case to find both the minimum and maximum of n numbers. 3
- (b) Define the NP and NP hard problem with examples. How many steps are required to prove that a decision problem is NP complete? 3
- (c) Is the following recurrence for knapsack problem is correct? If not, give the correct recurrence. Justify your answer in either case. 4
 If $w < w_i$ then $OPT(i, w) = opt(i - 1, w)$
 Otherwise, $opt(i, w) = \max(opt(i - 1, w), V_i + opt(i - 1, w))$